

An Inequality for Riemann Surfaces

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1. Introduction

This note contains a proof of an elementary but useful inequality for Riemann surfaces with a finitely generated non-Abelian fundamental group.

We recall that every Riemann surface S with a finitely generated fundamental group may be obtained from a closed Riemann surface of some genus $p \geq 0$ by removing $v \geq 0$ distinct points (called the *punctures* of S) and $\mu \geq 0$ disjoint Jordan domains with real analytic boundary curves (called the *ideal boundary curves* of S). The pair $(p, v + \mu)$ will be called the *topological type* of S , and the triple (p, v, μ) its *quasiconformal type*.

If S is not of quasiconformal type $(0, v, 0)$ with $v = 0, 1, 2$ or $(1, 0, 0)$, then S can be represented as the quotient U/Γ where U is the upper half-plane in the z -plane (viewed as the Poincaré model of the non-Euclidean plane) and Γ a torsion free Fuchsian group acting on U . In this case S carries a unique Poincaré metric (complete conformal metric of curvature (-1)) induced by the Poincaré metric

$$|dz|/\text{Im } z$$

in U . This is so, in particular, if the fundamental group of S is not finitely generated or, as we assume from now on, if S is of type (p, n) with

$$a = 2p - 2 + n > 0. \quad (1)$$

This inequality is equivalent to the requirement that the (finitely generated) fundamental group of S be non-Abelian.

A simple closed (Poincaré) geodesic on S will be called a *loop*, an *outer loop* if it can be deformed into an ideal boundary curve of S , and an *inner loop* if it cannot. There are as many outer loops as there are ideal boundary curves, and they are disjoint. The complement of an outer loop C has two components, one of the same quasiconformal type as S , the other (called the *funnel* belonging to C) of quasiconformal type $(0, 0, 2)$.

The complement of all funnels is the *Nielsen core* (also called Nielsen kernel) S_* of S . The relation

$$\text{area of } S_* = 2\pi a \quad (2)$$

follows from the Gauss-Bonnet theorem.

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Every Jordan curve on S can be deformed either into a point, or into a puncture, or into an ideal boundary curve, or into an inner loop. One can draw on S

$$d = 3p - 3 + n \quad (3)$$

(and no more) disjoint inner loops.

All statements made above are well known.

2. Statement of Results

Theorem 1. The length l of the shortest inner loop (if any) on S is bounded from above by a bound depending only on the number a and on the length L of the longest outer loop (if any).

A rather direct consequence is

Theorem 2. There are d disjoint inner loops on S whose lengths do not exceed a bound depending only on the number a and on the length of the longest outer loop (if any).

This estimate is of importance in the theory of moduli. It was conjectured by Mumford (unpublished) for $n = 0$. I announced the result in [2] but did not publish the proof; a sketch of the argument will be found in [1]. Recently Buser [3] gave another proof, for $n = 0$, and obtained some information about the order of magnitude of the bound involved.

Before proving the theorems we collect some needed lemmas. All refer to a Riemann surface S of type (p, n) satisfying (1), but actually hold, sometimes with obvious modifications, for all Riemann surfaces with a Poincaré metric.

3. Lemmas

Lemma 1. Assume that the complement of an inner loop C in S is connected. Then $S - C$ can be represented as a subdomain of a Riemann surface $\hat{S}$ of type $(p - 1, n + 2)$ so that

- (i) the Poincaré metrics on S and on $\hat{S}$ have the same restriction to $S - C$;
- (ii) the complement of $S - C$ in $\hat{S}$ consists of two outer loops on $\hat{S}$, each of the same length as C , and of the funnels belonging to these loops.

Proof. Let λ denote the length of C and let Γ be the Fuchsian group generated by $z \mapsto e^\lambda z$. On the Riemann surface $\Sigma = U/\Gamma$ there is a unique simple closed Poincaré geodesic B of length λ ; let F denote one of the two components of $\Sigma - B$. One constructs $\hat{S}$ by attaching, in an obvious way, a copy of $B \cup F$ to each bank of C . The details are left to the reader.

Lemma 2. Assume that the complement of an inner loop C in S has two components S_1 and S_2 . Then S_j is of topological type (p_j, n_j) with

$$a_j = 2p_j - 2 + n_j > 0, \quad j = 1, 2 \quad (4)$$

and

$$p_1 + p_2 = p, \quad n_1 + n_2 = n + 2, \quad a_1 + a_2 = a. \quad (5)$$

Also, S_j can be represented as a subdomain of a Riemann surface $\hat{S}_j$ of the same type as S_j so that

- (i) The Poincaré metrics on S and on $\hat{S}_j$ have the same restriction to S_j ;
- (ii) the complement of S_j in $\hat{S}_j$ consists of an outer loop on $\hat{S}_j$, of the same length as C , and of the funnel belonging to this loop.

The proof is similar to that of Lemma 1.

Let P be a puncture on S . A *collar about P* is a doubly connected domain on S bounded by P and a Jordan curve (called the boundary curve of the collar) orthogonal to the pencil of geodesics emanating from P . A collar about P of area β will be called a β -collar.

Lemma 3. The boundary curve of a β -collar about a puncture has length β . Any Jordan curve C which is freely homotopic to P and lies outside this collar has length at least β .

Proof. Represent S as U/Γ (cf. Sect. 1) and assume, since this can be achieved by conjugation, that Γ contains a primitive element $z \rightarrow z + 1$, and the canonical map

$$U \rightarrow U/\Gamma = S \quad (6)$$

takes vertical lines in U into geodesics emanating from P . This implies that a collar about P is the image under (6) of a region $0 \leq x < 1, y > \eta$ (for some $\eta > 0$) on which (6) is injective. One computes at once that, for a β -collar, $\eta = 1/\beta$.

The boundary curve is the one-to-one image under (6) of the segment $z = x + i/\beta, 0 \leq x < 1$, while a Jordan curve C having the properties stated in the lemma is the image of a Jordan arc joining a point z_0 on the imaginary axis to $z_0 + 1$ and lying in the strip $0 < y \leq 1/\beta$. A trivial calculation shows that the former has length β and the latter length at least β .

Let C_0 be a loop on S . A *collar about C_0* is a doubly connected domain on S bounded by two Jordan curves (the boundary curves of the collar) each point of which has the same distance from C_0 . A collar about C_0 of area 2β (sic) will be called a β -collar.

Lemma 4. The boundary curves of a β -collar about a loop C_0 of length λ_0 have length

$$\lambda = \sqrt{\lambda_0^2 + \beta^2}. \quad (7)$$

Any Jordan curve C which is freely homotopic to C_0 and lies outside this collar has length at least λ .

Proof. Represent S as U/Γ and assume, since this can be achieved by conjugation, that Γ contains a primitive element $z \rightarrow e^{\lambda_0} z$ and that C_0 is the image under (6) of the positive imaginary axis. This implies that a collar about C_0 is the image under (6) of a region

$$1 \leq |z| < e^{\lambda_0}, \quad |\arg z - \frac{\pi}{2}| < \theta_0$$

(for some θ_0 , $0 < \theta_0 < \frac{\pi}{2}$) on which (6) is injective. Introduce polar coordinates r , θ in U by setting $z = ire^{i\theta}$, $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. The Poincaré metric in U becomes $\sec^2 \theta [(dr^2/r^2) + d\theta^2]$. One computes that

$$\beta = \lambda_0 \tan \theta_0.$$

A boundary curve is a one-to-one image under (6) of the segment $1 \leq r < e^{\lambda_0}$, $\theta = \pm \theta_0$, while a Jordan curve C having the properties stated in the lemma is the image of a Jordan arc joining a point z_0 with $|z_0| = 1$ to a point $z_0 e^{\lambda_0}$ and lying outside the sector $|\theta| < \theta_0$. An easy calculation shows that the former has length

$$\lambda = \lambda_0 \sec \theta_0$$

and the latter length at least λ . The expressions obtained above for λ and for β imply (7).

Lemma 5. A β -collar, about a loop of length λ_0 or about a puncture (in which case we set $\lambda_0 = 0$) contains a γ -collar for every γ , $0 < \gamma < \beta$. Every point on a boundary curve of the former has distance

$$\log \frac{\beta + \sqrt{\beta^2 + \lambda_0^2}}{\gamma + \sqrt{\gamma^2 + \lambda_0^2}}$$

from the boundary of the latter.

The proof is left to the reader.

The following statement is known as the Collar Lemma.

Lemma 6. There is a positive continuous decreasing function $\alpha(t)$, $t \geq 0$, such that about every loop of length λ_0 there is a β -collar for every $\beta < \alpha(\lambda_0)$ and about every puncture there is a β -collar for every $\beta < \alpha(0)$. Two of these collars, about two disjoint loops, or about a loop and a puncture, or about two distinct punctures, are disjoint.

The best possible value of $\alpha(t)$ is

$$\alpha(t) = \frac{t}{2} \operatorname{csch} \frac{t}{2}. \quad (8)$$

The existence of $\alpha(t)$ for small t has been proved by Keen [5], for all t by Halpern [4]; the sharp formula (8), and an extension to Fuchsian groups with torsion are due to Matelski [7], cf. also Randol [8], Buser [3]. The assertions concerning punctures are contained in the work of Shimizu [9] and Leutbecher [6].

Note that in view of Lemmas 1 and 2 it suffices to prove Lemma 6 for a Riemann surface of type $(0, 3)$; this case can be handled by elementary non-Euclidean geometry.

4. Proof of Theorem 1

There is nothing to prove if $d = 0$ [cf. (3)] which happens only for $(p, n) = (0, 3)$. Assume that $d > 0$, and denote by L the length of the longest outer loop of S ; if there are no outer loops, set $L = 0$. Set

$$2\beta = 3\gamma = \alpha(L), \quad \delta = \log \frac{\beta + \sqrt{\beta^2 + L^2}}{\gamma + \sqrt{\gamma^2 + L^2}}$$

where $\alpha(t)$ is the function in Lemma 6, and

$$2\varepsilon = \min(\gamma, \delta).$$

Then $\varepsilon > 0$ and depends only on L .

Let $B_1, \dots, B_n$ be the β -collars and $B'_1, \dots, B'_n$ the γ -collars about the punctures and outer loops of S . Then $B'_j \subset B_j$ and the $B_1, \dots, B_n$ are disjoint (by Lemma 6). Let S_{**} be the complement of $B_1 \cup \dots \cup B_n$ in the Nielsen core S_* of S (i.e. $S_{**} = S_* - (B_1 \cup \dots \cup B_n)$; if $n = 0$, then $S_{**} = S_* = S$). Since $\varepsilon \leq \delta$, Lemma 5 implies that

$$\text{the } \varepsilon\text{-neighborhood of } S_{**} \text{ lies in } S_*. \quad (9)$$

If there is on S an inner loop of length not exceeding 2ε , the assertion of Theorem 1 is valid for S , the desired bound being 2ε . We assume there is no such loop. Then

$$\text{every point in } S_{**} \text{ is the center of a (Poincaré) disc on } S \text{ of radius } \varepsilon. \quad (10)$$

Indeed, assume a point Q in S_{**} does not have this property. Then there is a Jordan curve C , beginning and ending at Q , which consists of two geodesic segments, each of length ε' , $0 < \varepsilon' \leq \varepsilon$. This C lies in S_{**} .

Represent S as U/Γ (cf. Sect. 1) and lift C to a point $z_0 \in U$ which is mapped by (6) onto Q . We obtain a Jordan arc with endpoints z_0 and $z_1 \neq z_0$, and there is a $g \in \Gamma$ with $g(z_0) = z_1$. If g were parabolic, C could be deformed into a puncture, but this is impossible by Lemma 3, since $2\varepsilon \leq \beta$ and C is of length $2\varepsilon'$. Thus g is hyperbolic and C can be deformed into a closed geodesic C_0 of length not exceeding $2\varepsilon' < 2\varepsilon$ (the image of the axis of g under (6)). Since C is a Jordan curve, so is C_0 . By Lemma 4, C_0 cannot be an outer loop, since $2\varepsilon' < \beta$. Thus C_0 is an inner loop, but this contradicts our assumptions. Hence (10) holds.

Assume now that the genus p of S is positive. Then there are closed curves on S which are not homologous to 0 modulo the boundary of S ; call such curves essential. There must be essential inner loops, and among them one, call it C_0 , of minimum length λ_0 . By Lemma 6, C_0 lies in S_{**} .

Set $m = [\lambda_0/3\varepsilon]$. There are points $Q_1, \dots, Q_m$ on C_0 such that the length of an arc of C_0 between any two of them is at least 3ε . We claim that the distance between any two exceeds 2ε . Indeed, if it were not so, the cycle represented by C_0 would be the sum of two cycles, each represented by a closed curve of length less than λ_0 . At least one of these curves would have to be essential, and this would imply the existence of an essential inner loop of length $\lambda_1 < \lambda_0$, contradicting the extremal property of C_0 .

By (10) every point Q_j is the center of a disc of radius ε and thus of area $4\pi \sinh^2(\varepsilon/2)$. By what was proved above these discs are disjoint, and by (9) they lie in S_* . Recalling (2) we conclude that

$$m \leq \frac{a}{2} \operatorname{csch}^2 \frac{\varepsilon}{2}.$$

Since

$$\lambda_0 \leq 3\varepsilon(m+1)$$

by the definition of m , and since λ_0 is an upper bound for the length l of the shortest inner loop, the assertion of Theorem 1 is verified in the case considered, i.e. for $p > 0$.

Assume next that $p = 0$. Note that $n > 3$, cf. (2) and (3), and recall that $d > 0$. Now S is a plane domain of connectivity n , and S_{**} a subdomain of the same connectivity, bounded by the disjoint Jordan curves $K_1, \dots, K_n$ (boundary curves of the β -collars $B_1, \dots, B_n$). By Lemmas 3 and 4,

$$\text{length of } K_j \leq \sqrt{\beta^2 + L^2}, \quad j = 1, \dots, n;$$

this bound depends only on L .

A simple argument shows that among all curves in S joining a K_i to a K_j , $i \neq j$, $1 \leq i, j \leq n$, there is at least one, call it C_0 , of minimum length λ_0 . This C_0 cannot intersect any of the curves $K_1, \dots, K_n$ for otherwise it could be replaced by a shorter one. Hence it lies in S_{**} . It is clearly a geodesic arc. For the sake of definiteness assume that it joins K_1 to K_2 . Reasoning exactly as before we obtain a bound for λ_0 depending only on a and L .

There is a Jordan curve C_1 which separates $B_1 \cup B_2$ from $B_3 \cup \dots \cup B_n$, lies arbitrarily close to $C_0 \cup K_1 \cup K_2$ and has length arbitrarily close to

$$\lambda_1 = 2\lambda_0 + 2\sqrt{\beta^2 + L^2}.$$

Hence there is an inner loop on S of length not exceeding λ_1 . Since $l \leq \lambda_1$, the assertion of Theorem 1 holds also for $p = 0$.

5. Proof of Theorem 2

We assume, without loss of generality, that the bound for l in Theorem 1 increases with a and with L .

All ordered pairs of nonnegative integers p and n satisfying (1) can be arranged into a sequence by requiring that $(p' n')$ precede (p, n) if and only if either

$$a' = 2p' - 2 + n' < 2p - 2 + n = a$$

or

$$a' = a \quad \text{and} \quad p' < p.$$

The first term of the sequence is $(0, 3)$; for Riemann surfaces of this type Theorem 2 is trivially true. We prove it for a Riemann surface S of type (p, n) assuming it has been established for all preceding types.

By Theorem 1, there is on S an inner loop C whose length l can be estimated in terms of a and L , the length of the longest outer loop on S .

Assume first that $S - C$ is connected. As in Lemma 1, we identify $S - C$ with a subdomain of a Riemann surface $\tilde{S}$ of type $(p - 1, n + 2)$. This type precedes (p, n) . The length of the longest outer loop on $\tilde{S}$ is $L' = \max(l, L)$. By the induction hypothesis there are

$$d_1 = 3(p - 1) - 3 + n + 2 = d - 1$$

disjoint inner loops on $\tilde{S}$ whose lengths can be estimated in terms of a and L .

Assume next that $S - C$ has two components, S_1 and S_2 . As in Lemma 2, we identify S_j with a subdomain of a Riemann surface $\tilde{S}_j$ of type (p_j, n_j) , $j = 1, 2$. In view of (4) and (5), the type (p_j, n_j) precede (p, n) . The length of the longest outer loop on S_j does not exceed $\max(l, L)$, so that, by the induction hypothesis, there are

$$d_j = 3p_j - 3 + n_j$$

disjoint inner loops on $\tilde{S}_j$ (and actually on $(\tilde{S}_j)_*$ and thus on S_j) whose lengths can be estimated in terms of $\max(l, L)$ and a_j and hence also in terms of L and $a_1 + a_2 = a$. Together with C we obtain

$$d_1 + d_2 + 1 = d$$

disjoint inner loops on S with the required property.

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